



MID SEMESTER EXAMINATION, AUTUMN-2024

School of Applied Sciences, KIIT-DU

Course: Basic Electrical Engineering

Code: EE10002

Full Marks: 20

Time: 90 minutes

Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

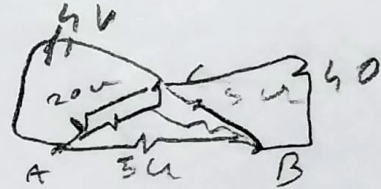
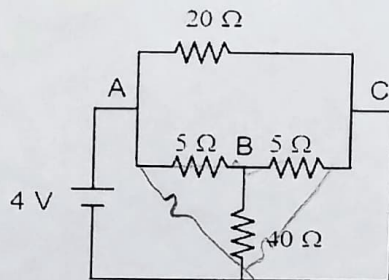
All parts of a question should be answered at one place only.

Q.1

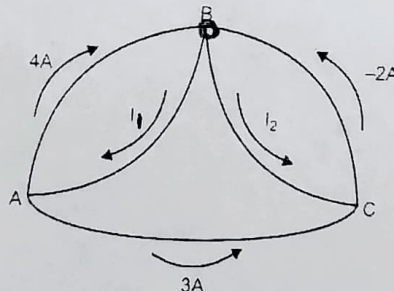
- modification*
- | | Marks | CO |
|---|-------|-----|
| a) Define Bilateral and Unilateral element with one example in each case. | 1 | CO1 |
| b) What is the difference between mesh and loop in an electrical circuit? | 1 | CO1 |
| c) If a 3 ohm, 4 ohm, 5 ohm resistors are connected in delta. Find its equivalent star resistances? | 1 | CO1 |
| d) What is the difference between DC and AC? | 1 | CO2 |
| e) Define the following terms (i) Time Period (ii) Phase of an alternating quantity. | 1 | CO2 |

Q.2

- | | Marks | CO |
|--|-------|-----|
| a) Illustrate the current delivered from the 4V battery in the circuit shown below using delta to star conversion. | 3 | CO1 |



- b) Determine the values of currents I_1, I_2 by using KCL in the circuit shown below.

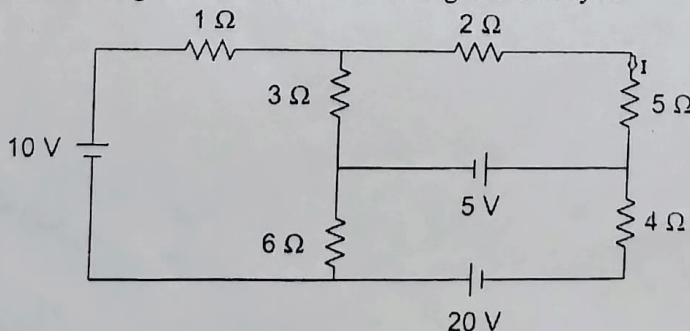


$$\begin{aligned} 4A - 2A - I_1 - I_2 &= 0 \\ -4A - 3A + I_1 &= 0 \\ 2A \end{aligned}$$

Q.3

- a) Find the current flowing in the 5 ohm resistor using mesh analysis?

Marks	CO
3	CO1



P.T.O